BBO capstone project presentation template

Develop a concise presentation that captures:

1. **An overview of your BBO approach**
   - Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
   - Briefly describe your overall strategy and how your process fits together.

Objective: maximize eight unknown black-box functions with a limited sequential query budget and learn where the strongest regions are over time.

Overall approach:
- start with exploratory queries to map the search space
- fit surrogate models to observed data after each round
- score candidates using predicted value, uncertainty, and novelty
- adapt the balance of exploration and exploitation by function and by dimension

The process is iterative: each submission updates my local evidence, changes the surrogate view of the search space, and informs the next round. I also keep scripts, reports, and reflections so the strategy is reproducible rather than purely intuitive.

2. **How your strategy has evolved**
   - Describe the key changes in your approach since your early rounds.
   - What influenced these changes – data trends, feedback, model performance or intuition?
   - Identify which principles or heuristics now guide your query decisions.

My strategy has changed substantially since the early rounds. At first, the process was broad and heuristic, with simple uncertainty-aware exploration. As more data arrived, I moved to tuned surrogate models, ensemble uncertainty estimates, hybrid scoring rules, and later clustering/PCA-style structure analysis.

These changes were driven by:
- repeated data trends, especially strong boundary behavior in some functions
- feedback from model fit and cross-validation
- the need to reduce ad hoc choices and make the process easier to explain

The principles that now guide my decisions are: trust repeated structure more than isolated points, use uncertainty as a guardrail rather than the only objective, exploit stable low-dimensional patterns sooner, and keep more exploration in sparse high-dimensional functions.

3. **Patterns, data and insights**
   - Highlight the most meaningful trends you've observed across your data points or model evaluations.
   - Identify which variables or behaviours seem to influence results the most.
   - Explain how these observations have shaped your understanding of the search process.

The strongest patterns in my data are not uniform across functions. Functions 1, 2, and 5 often reward boundary-adjacent regions, while Function 4 behaves more like a basin with neighboring pockets. In higher-dimensional functions, only a subset of variables seems to drive most of the useful variation.

That led me to look for:
- repeated local neighborhoods rather than one-off high points
- dominant directions in the data, similar to principal components
- evidence that some dimensions contribute far more than others

These observations changed my understanding of the search process. Instead of treating the space as equally important everywhere, I now view it as a mixture of ridges, basins, and sparse unresolved regions that need different treatment.

4. Decision-making and iteration

- Explain how you balance exploration and exploitation.
- Describe one or two examples of strategic decisions. What worked, what didn't, and why?
- Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

I now balance exploration and exploitation with an explicit scoring rule that combines predicted value, uncertainty, novelty, and local structure.

Examples of strategic decisions:
- What worked: exploiting repeated boundary or centroid patterns when several rounds supported the same region
- What did not work: over-chasing novelty, relying on one surrogate family everywhere, or overreacting to one surprising point

When results do not match expectations, I do not just repeat the same move. I retrain or retune the model, inspect influential dimensions, and shift the next query mode between exploit, balanced, and explore. That makes the process more resilient when the black-box behavior is unstable or only partially observed.

5. **Next steps and reflection**
   - Outline your planned actions or experiments to improve the model's performance.
   - Include two or three sentences connecting your work to the broader ML landscape.
   - Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

Next steps:
- update the local arrays with the latest portal outputs
- rerun the latest generator on the full data state
- use the final round to exploit the strongest recurring directions while preserving a small amount of exploration in unresolved high-dimensional functions

Broader ML connection: this project reflects real optimization workflows in machine learning, including surrogate modeling, uncertainty management, interpretability, and iterative decision-making under limited feedback. The same ideas appear in hyperparameter tuning, experiment design, and human-in-the-loop optimization.

For a non-technical audience: I would explain that I am using each test result to learn where the best settings are likely to be, then choosing the next test in a way that balances promising areas with a small amount of careful exploration.